

# Concurrent Resistance Training Enhances Performance in Competitive Distance Runners: A Review and Programming Implementation

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## ABSTRACT

Concurrent training, defined as resistance training sessions programmed alongside endurance training sessions during the same cycle, is a tool that highly trained runners can use to augment performance. A review of the current literature explains how resistance training focused on increasing maximal force development and power output can improve running economy, a key performance indicator for endurance running. This review explains the current concepts in concurrent training for competitive distance runners, discusses misconceptions related to concurrent training, and offers practical implementation examples for coaches, athletes, and strength and conditioning practitioners working with this special population.

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## INTRODUCTION

Concurrent resistance and endurance training have many performance benefits for distance running athletes that are supported by the literature. Progressive resistance training is a tried and true method of increasing maximal strength, rate of force development, and muscular power (3,36). Each of these qualities are beneficial for the distance runner; however, many running athletes have not implemented concurrent training into their programming due to a lack of knowledge or misconceptions. Concurrent training, defined as resistance training sessions programmed alongside endurance training sessions during the same cycle, is a tool that highly trained runners can use to augment performance (36). This review focuses on the competitive distance running population, operationally defined as both female and male athletes competing in 10,000 meters to

marathon distance races with greater than  $55 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} \cdot \dot{V}\text{O}_{2\text{max}}$ , indicating well-trained and highly trained runners (13). Recreational runners and competitive sprinters have programming and periodization needs that are outside the scope of this review. The purpose of this review is to explain the current concepts and literature on concurrent training for competitive distance runners, discuss misconceptions related to concurrent training, and offer practical implementation examples for coaches, athletes, and strength and conditioning practitioners working with this special population.

## DISTANCE RUNNING PERFORMANCE INDICATORS

Key performance indicators in competitive distance running are referred

### KEY WORDS:

distance running; concurrent training; resistance training; running economy; periodization; exercise prescription

to as running economy (RunE), maximal oxygen uptake ( $\dot{V}O_{2\max}$ ), and lactate threshold (LT) (2,21). Research demonstrates positive  $\dot{V}O_{2\max}$  and LT training effects due to concurrent training in recreational and generally sedentary individuals; however, there is minimal research to support large or meaningful adaptations in either indicator with concurrent training in the highly trained runner due to an insufficient stimulus (2,13). Although minimal positive adaptations to  $\dot{V}O_{2\max}$  or LT are directly suggested with concurrent training for elite runners, research demonstrates there are no negative effects on these performance indicators in trained endurance athletes (21). Therefore, most clinical trials experimenting with performance enhancement for runners focus on RunE, the key performance indicator that may yield meaningful positive change.

Indeed, positive adaptations to RunE are consistently demonstrated in clinical trials with the prescription of concurrent resistance training (6,10,32,41). RunE is defined as the metabolic cost or energy demand for a given velocity of submaximal running to cover a given distance (5). Jung describes that a runner requiring a lower  $\dot{V}O_2$  at the given velocity has better economy. Improved running economy allows a runner to run faster over a given distance or to run longer at a constant speed due to reduced consumption of oxygen (21). In a recent systematic review including middle- and long-distance running athletes, duathletes, and triathletes performing concurrent high-load resistance training ( $>80\%$  1 repetition maximum [RM] or isometric 60–90% 1RM), RunE improved 2–8% with an effect size of 0.14–3.22. The author states the typical error of measurement of running economy ranges from 1 to 2%, and the smallest worthwhile change is approximately 2%. A 2% improvement is believed to represent a real improvement and is not due to variability of the measure. It is likely that the improvements in running economy with concurrent training in the 2–8% range do represent

a meaningful change in running performance (10).

Storen et al. found improvements in RunE without changes in body weight or  $\dot{V}O_{2\max}$  after 8 weeks of strength training in 17 well-trained distance male and female runners. Running training was performed as usual, and the intervention group added 4 sets of 4RM half-squats with 3-minute rest for 3 sessions per week with progressive overload using free weights (41). Repetition maximum is defined as the greatest amount of weight that can be lifted with proper technique for only one repetition or for a specified number of repetitions (3). Another study, performed by Sedano et al. on 18 highly trained male runners, explored the effects of RunE after 12 weeks of resistance training plus explosive strength training at high loads (3 sets of 7 reps at 70% 1RM) versus resistance training at low loads (3 sets of 20 reps at 40% 1RM) and very low loads (3 sets of 25 reps of elastic band exercises). This author found consistently greater improvements in RunE, peak velocity, and rating of perceived exertion in the high-load resistance training group and the low-load resistance training group over the very low-load resistance training group. No change in body mass was reported in any group (37).

Positive adaptations in RunE are likely influenced by several physiological and biomechanical factors (4,8,17). These physiological adaptations include increases in muscle cross-sectional area and absolute strength, which contribute to distance running at a lower relative intensity (17), recruitment of lower threshold motor units in submaximal running efforts (4,8), and increased Achilles tendon stiffness (16). Alterations in biomechanical factors such as stride length and foot position at initial contact may also play a role in RunE (17).

## CONCURRENT TRAINING FOR DISTANCE RUNNERS

Exercise selection and the prescription of concurrent training must be designed to meet the unique needs of the distance

runner. Many clinical trials prescribe high-load and low-repetition resistance training, traditionally referred to as strength training and power training, to induce improvements in RunE. Contrary to outdated clinical practice performing muscular endurance resistance training (weight training featuring low-load, high-repetition, and short-rest intervals), strength and power resistance training result in greater benefit to the endurance athlete.

Strength training, traditionally defined as resistance training greater than or equal to 85% 1RM load, 2–6 repetitions, with at least 2-minute rest intervals (3) that target maximal force development, increases the rate of force development and tendon stiffness (24,25), which is correlated with improved RunE by reducing shortening-induced energy cost (16,34). Explosive strength exercises are prescribed heavily in the literature to increase RunE, maximal strength development, and peak power measurements (6). Explosive strength exercises are described as medium to high load (75–95% 1RM), high velocity plyometrics (6). In addition, research suggests that reactive strength exercises such as low-load, high-velocity plyometrics enhance rate of force development by developing stretch-shortening cycle utilization and leg musculotendinous stiffness (6,34,40). Exercise selection based on these study protocols for maximum strength development, explosive strength, and reactive strength improvements are denoted in Table 1 and described below in this review.

According to a review by Blagrove et al., (10) the training variables that have the greatest impact on RunE include the prescription of multi-joint free-weight exercises as compared to machine and single-joint exercises, exercises with progressive and sufficient overload, exercises specific to the task of running, and increased frequency of resistance training sessions.

Complex resistance training is a method of alternating 2 modes of exercise prescription in one training session that may be an efficient

**Table 1**  
**Recommended exercise selection for competitive distance runners to induce adaptations in running economy (4,10,36)**

| Training goal                                                        | Exercise selection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Maximal strength development                                         | <p>Deadlift and Romanian deadlift variation—BB conventional or sumo DL, DB conventional or sumo DL, BB RDL, DB RDL, and single-leg deadlift</p> <p>Back squat variation—BB back squat, DB or KB goblet squat, box squat, BB front squat, machine leg press and single-leg squat</p> <p>Split-squat variation—unweighted split squat, walking lunge, BB or DB split squat, Bulgarian split squat, front-raised split squat, and deficit split squat</p> <p>Plantarflexion variations—double leg, single leg, on floor, on raised platform, weighted with DB, weighted with BB on back</p> <p>Hamstring curl—Nordic hamstring, machine hamstring curl</p> <p>Knee extension—machine knee extension</p> |
| Explosive strength development                                       | <p>Clean or rack pull</p> <p>High-effort box jump</p> <p>Countermovement jump</p> <p>Jump squats—BB 1/4 jump squats and unweighted jump squats</p> <p>Overhead push press</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Reactive strength development                                        | <p>Bounding</p> <p>Pogo jumps</p> <p>Drop or depth jumps</p> <p>Tuck jumps</p> <p>Hurdle hops</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| BB = barbell; DL = deadlift; DB = dumbbell; RDL = Romanian deadlift. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

programming option for athletes. Li et al. (26) investigates complex resistance training by alternating heavy resistance strength exercises and explosive strength exercises within a given concurrent resistance training session and compared results with a heavy resistance training group and a strength-endurance training group. All runners in each group performed resistance training 3 times per week and endurance running sessions 9 times per week. The complex training group used pairs of exercises alternating between multi-joint, high-load closed-chain exercise (80–85% 1RM, 3 sets of 5–6 repetitions of back squat, Bulgarian squat, and Romanian deadlift) and medium-to-high load plyometric training (3 sets of 5–6 repetitions of drop jump from 40-cm box, single-leg hop, and a double-leg 50-cm hurdle hop) with an intracomplex rest interval of 4 minutes. The heavy resistance training group used the same resistance exercises and intensity, excluded the plyometrics, and added 2 sets of each resistance exercise to balance overall

volume load. The strength-endurance group used the same resistance training exercises with a decrease in intensity to 40% 1RM, an increase in repetitions to 20–30, excluded the plyometrics, and increased the sets by 2 to balance overall volume load. Greatest improvements in RunE, strength and power output, plyometric reactive strength, and timed running performance were found in the complex training group, followed by the heavy resistance training group (26).

It should be noted that the interference effect is an important consideration in concurrent training. The interference effect explains that the adaptation to endurance or strength performance may be hindered by the participation in the other form of training (36). One mechanistic explanation of the acute interference effect is that a long running training session blunts the protein synthesis response needed for strength and hypertrophy gains for 3–12 hours after exercise, therefore compromising optimal strength development (36). Because of the high number of weekly endurance

running sessions programmed for the competitive distance runner (between 6 and 9 running sessions per week depending on the periodization phase), there may be insufficient intersession recovery time between running sessions and resistance training sessions to fully realize the anabolic stimulus from resistance training. This acute interference effect may be avoided with appropriate intersession recovery programming that will be discussed later in this review. Because of the lack of homogeneity of study protocols and the lack of protocols for longer periods, there are no scientific conclusions about the mechanism of the chronic interference effect of concurrent training.

## CONCURRENT TRAINING MISCONCEPTIONS

Anecdotal misconceptions of concurrent training that are not supported in the literature often deter runners from stepping foot in the gym. The belief that negative adaptations in body composition will result from concurrent training is a common misunderstanding that is not supported by the literature at large.

Weight gain, body composition changes, and limb girth measurements do not change significantly in male and female distance runners performing concurrent training (10). The incorrect belief that resistance training will increase muscle mass and lead to a slower race pace is based on unsubstantiated claims and anecdotal tradition. An observational study of 341 marathoners and ultramarathoners in 2015 examined mean training distance, mean training pace, mean race pace, and several anthropometric factors from a large cohort of marathoners and 100-km ultramarathoners from 3 separate investigations (only one female subject included in total subject count) (42). No associations between body fat and race pace were found in runners if their body fat measured below 15%, indicating that body composition for distance runners competing in 42- to 100-km races seem to be of less importance if the runners have a relatively low level of fat mass. This study suggests that it is advantageous to be a runner with a high proportion of lean muscle mass as compared to body fat (42). Logically, this trend may be extrapolated to female runners as well; however, the specific advantageous body composition cutoff numbers for female runners are not clear at this time. In addition, a review of RunE written from a muscle energetic perspective provided similar conclusions, indicating that runners with a high proportion of skeletal muscle mass and low fat mass may have an advantage in reducing the absolute energy cost of running (17). Using a lower energy cost in a race contributes to less heat generation and a lower relative muscle activation needed for running at a given speed contributing again to improvements in RunE (17).

Although higher proportions of fat-free body mass and lower proportions of fat mass are advantageous to runners, there is very little evidence indicating resistance training increases gross body mass or body weight in this population (28). It is possible that hypertrophic muscle mass increases

were not identified due to the generally short-term duration of studies. There is a possibility that small increases in proximal muscle groups such as the gluteals and quadriceps would be recognized with longer training durations but may not result in an overall change in gross body composition. Two recent systematic reviews on concurrent training generally found negligible changes in body composition measurements but still reported improvements in running performance in both female and male runners (2,10). Notably, most research studies are short-term in duration, generally less than 10 weeks, and the chronic effects of concurrent training are less clear. In endurance athletes, the acute interference effect lessens the effects of resistance training on muscular strength, muscular power, and muscle mass, which may explain the absence of body composition changes (36). Thus, with appropriate programming of concurrent training, the addition of strength-focused resistance training sessions for a competitive distance runner may result in positive adaptations to skeletal muscle (and the system), induce changes to RunE, and result in no gross body composition changes.

## PROGRAMMING APPLICATION OF CONCURRENT TRAINING FOR DISTANCE RUNNERS

### EXERCISE SELECTION

Most concurrent training studies to date in this unique population implement closed-chain, high-load, low-velocity multi-joint exercises using free weights, body weight, or resistance machines for maximal strength development. The most frequently used exercises include the back squat and leg press variations (6,7,9,12,15,18,22,28,32,34,35,38,41,43), Romanian-style deadlift or deadlift variations (6,22,38), plantarflexion variations (1,12,15,16,22,32,35,43), split-squat variations (6,18,22,32,35), hamstring curl (15,28,34,35), and knee extension (12,15,28,32) (Table 1).

Very few studies included upper-body exercises in their investigations (15,32,35), although practically some

power exercises programmed to increase the rate of force development of hip extension may require some level of upper-body strength including push press and clean pull variations.

Although deadlift variations were not as popular in the literature, applying the perspective of injury risk reduction and posterior chain maximal power output development, it would be prudent to use this exercise in at least one session per week for distance running athletes.

Multi-joint exercises are a time-efficient method for maximal strength and power gains due to multiple large muscle group activation with a singular closed-chain exercise. Applying concepts in the strength and conditioning literature and understanding exercise prescription, if time is limited, it is advised to implement complex multi-joint exercises such as a squat, a deadlift, and split-squat variations and exclude accessory exercises and plyometrics for that session.

The most commonly reported plyometric exercises were squat jumps (6,11,28–30,35), countermovement jumps (6,11,30,34), drop or depth jumps from a height (6,30,33,35), and variations of jumps, bounds, and hurdle hops (6,11,18,29–31,34,35,40,42) (Table 1). Many studies did not report the intensity or rating of perceived exertion goal for their plyometric exercises, and therefore, differentiation of explosive strength versus reactive strength exercise selection recommendations are limited.

### EXERCISE INTENSITY AND VOLUME

The most common prescription of intensity, or training load, for concurrent resistance training exercises with the goal of maximal strength development ranged between 2 and 6 sets of 3–10 repetitions per exercise at loads greater than 70% 1RM (10). Applying the principles of training prescription, it would be sensible to recommend that the strength and conditioning practitioner implement methods of estimating 1RM on a regular basis, tracking overall



tonnage to limit the effects of overtraining, and determining the criteria for load progression to reach optimal gains.

Common plyometric training prescription consisted of 1–6 exercises performed over 1–6 sets of 4–10 repetitions, averaging 30–228 “foot contacts” per session (10). Low training volume for plyometrics (60–72 contacts per session) may be sufficient for distance runners who may not be accustomed to such exercises (33). It is suggested to progress up to 110–228 foot contacts per session (10). Similarly, it would be sensible to track plyometric contacts per week and to consider the residual local and global fatigue effect of intensity between explosive strength and reactive strength plyometrics.

### **FREQUENCY**

The most common prescription of resistance training frequency in clinical trials is between 2 and 3 sessions per week (10). Depending on the training cycle, the ratio of running sessions to resistance training sessions per week may vary. It is suggested that improvements in RunE are observed in running to resistance training ratios between 6:2 and 9:3, otherwise stated in percentages of resistance training to overall weekly sessions, or 25% in this example (4). Greater frequency of resistance training sessions is advocated during the general preparatory phase (GPP) as compared to the specific preparatory phase (SPP) with periodized concurrent training (Table 2).

### **INTERSESSION RECOVERY**

Another consideration for periodization is the successive timing of resistance training sessions to running sessions. The acute interference effect describes how the residual fatigue from the first bout of exercise may decrease the performance of the subsequent bout. Therefore, maintaining a 4- to 6-hour window of intersession recovery between concurrent training sessions may alleviate the potential negative effects of subsequent sessions due to lessening the effect of reductions in neuromuscular function and strength that are evident for up to 24 hours after a heavy training session (39).

Damo et al. (14) determined that exercise session order in a single day is also an important consideration to reap the greatest benefits of each training session. In this study, a greater decrease in muscle voluntary contraction was noted when a resistance training session preceded a running training session by 6 hours versus when a resistance training session followed a running training session by 6 hours. The author concluded that the possible residual fatigue effects of the running session were eliminated and did not affect the performance of the resistance training session. Thus, expanding the rest period between sessions and ordering the sessions to perform running in the first session of the day may allow for greater positive adaptations to RunE.

### **PRACTICAL PERIODIZATION IMPLEMENTATION**

#### **CONCURRENT RESISTANCE TRAINING PERIODIZATION**

Periodization is a concept that many running coaches and athletes use to divide the year into training blocks and cycles to peak their performance on a specific date or dates (3). The individual running sessions will vary in pace, distance, and incline to peak the athlete for their most important races in the year. The addition of concurrent training programming should take running training sessions into consideration; however, resistance training periodization will follow its own unique variations in volume, intensity, and specificity. As a strength and conditioning professional working with distance runners, it is important to understand the goals of the individual, their resistance training age, and how to work with the running coach to create a feasible and intelligent plan. With careful programming, the athlete can add a novel training stimulus to their training program, such as resistance training, without changes in muscular fatigue and soreness.

Strength and conditioning professionals, along with health care practitioners prescribing exercise, are recommended to consider the runners' perceived physical

readiness on a daily basis (36). Practically, the athlete may report levels of soreness or overall increased fatigue that should be monitored and planned appropriately by adjusting caloric proportions, ensuring appropriate sleep, and monitoring stress. To determine the best plan for your athlete, a healthy discourse of mental state and physical readiness between practitioner and athlete must take place at every training session. Tracking training load using internal workload indicators, such as rating of perceived exertion, is a useful measure that the coach can use to objectively quantify a subjective state of psychological and physiological readiness (20).

The experienced health care practitioner and strength and conditioning professional recognizes the differences in athletes' motor control and their ability to perform general movement patterns. Although a runner may have years of experience with endurance racing, resistance training may still be a novel stimulus, and exercise selection considerations for this athlete may differ from a runner with more experience in resistance training. It is recommended to retain the variation in the recommended movement patterns (e.g., squat, hinge, and unilateral exercise) while adapting it to meet the needs of the individual athlete. Low training age describes the athlete in which resistance training is a novel stimulus, they perform the pattern with compensations or substitutions and require moderate or high feedback to perform it independently. Usually, these athletes have less than 1 year of quality resistance training experience. Athletes with a high training age will be able to perform the movement patterns with minimal to no feedback, demonstrate minimal to no compensations, and usually have greater than 1 year of quality resistance training experience. Detailed exercise selection based on the athlete's training age and phase of periodization is recommended in Table 3.

A competition schedule for competitive runners may include more than 15 races per year between training races with the goal of evaluating

| <b>Table 2</b><br><b>Recommended weekly frequency of concurrent training and running training intensity distribution in a periodized macrocycle (10,23,36)</b>                                                                                                             |                                               |                                                |                                                |                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------|------------------------------------------------|------------------------------------------|
|                                                                                                                                                                                                                                                                            | General preparatory phase                     | Specific preparatory phase                     | Maintenance phase                              | Taper/peak phase                         |
| Concurrent training frequency                                                                                                                                                                                                                                              | 4:2–5:3 (33–38%) <sup>a</sup>                 | 6:2–9:3 (25%) <sup>a</sup>                     | 6:1–2 to 9:2 (14–25%) <sup>a</sup>             | 4:1–2 to 5:2 (20–33%) <sup>a</sup>       |
| Running training intensity distribution                                                                                                                                                                                                                                    | Zone 1: 80%<br>Zone 2: 0–5%<br>Zone 3: 15–20% | Zone 1: 80%<br>Zone 2: 10–15%<br>Zone 3: 5–10% | Zone 1: 80%<br>Zone 2: 10–15%<br>Zone 3: 5–10% | Zone 1: 80%<br>Zone 2: 15%<br>Zone 3: 5% |
| Recommendations for weekly frequency of concurrent training represented as a ratio of running sessions to resistance training sessions. Running training intensity distribution is represented as percentages of weekly running training volume based on race-paced zones. |                                               |                                                |                                                |                                          |
| <sup>a</sup> Percentages represent resistance training sessions to total training sessions per week.                                                                                                                                                                       |                                               |                                                |                                                |                                          |

training variables and approximately 3–5 races per year that will require true taper and peaking phases. Commonly, the year may be broken down into 3–5 macrocycles to taper for these races, each composed of mesocycles consisting of the GPP, the SPP, the maintenance phase, and the tapering and peaking phase (34).

The GPP for distance running is similar to the off-season phase in team sports. In this phase, the volume of resistance training is high, and the intensity is moderate to high. Variability in exercise selection, as well as variability in running sessions, will be higher in the beginning and then taper as the phase progresses forward. The goal of this phase is to induce neuromuscular adaptations to new skills and movement patterns and then transition to build muscular strength (36). As the GPP transitions to maximal strength development, the volume of resistance training decreases and the intensity increases. This is also the phase that physical therapists rehabilitate nagging injuries. Resistance training sessions are generally scheduled for 2–3 sessions per week with running sessions scheduled for 5–6 sessions per week. Applying existing research and principles of training prescription, it is advised to program a greater percentage of resistance training sessions to overall weekly training sessions in the GPP (~30–40%) and a lower percentage of resistance training sessions to weekly sessions in the SPP (~20–30%) (Table 2).

Next, the athlete progresses into the SPP, similar to the preseason phase in team sports. Resistance training volume progressively decreases, intensity progressively increases, and the variability of training decreases. In this phase, running training distances and pace transition to resemble the competitions ahead. For example, a marathon runner will complete a greater number of runs more than 15 miles and closer to race pace. SPP is the phase to capitalize on the strength gains made in the GPP and transition to focus on explosive strength exercises to target changes in the rate of force development. Rehabilitation of injuries should be complete entering into the SPP, running training resumed if returning from injury, and new skill acquisition can be capitalized on. Resistance training sessions are generally scheduled for 2 sessions per week with running training sessions increasing up to 9 sessions per week.

As the runner progresses toward the tapering and peaking phase of his or her macrocycle, resistance training will enter a maintenance phase in which the volume will decrease, while intensity remains high. During the maintenance and tapering/peaking phases, resistance training sessions generally decrease to 1–2 sessions per week with significant decrease in overall volume to approximately 50% of overall tonnage and 50% of plyometric contacts compared with the previous SPP cycle (10).

Finally, rest and recovery may be programmed in the form of an active recovery phase after a maximum effort race to decrease the effects of mental fatigue and avoid the risks of overuse injury by overtraining. Another recovery option is the implementation of a deload week, a planned 4- to 7-day decrease in overall volume or intensity for the same purpose of an active recovery phase that occurs every 4–6 weeks of training in the GPP or SPP phases. If complete rest from all resistance training is programmed during the active recovery phase, then a week or 2 of introductory resistance exercise to gradually increase volume may attenuate muscular fatigue before starting a GPP due to the repeated bout effect (19). Table 4 outlines exercise prescription dosage recommendations for concurrent resistance training sessions while mindfully considering periodization for competitive distance runners.

## CONCURRENT ENDURANCE RUNNING PERIODIZATION

For most study protocols, running training sessions were completed as usual. In application, strength and conditioning practitioners may choose to not alter endurance training intensity or volume with the addition of concurrent training. Applying periodization principles, the intensity and volume of running training should undulate in a structured manner with resistance training according to the training block or cycle. For example, during the GPP, running sessions will

**Table 3**  
**Exercise selection recommendations for resistance training sessions per day of the week for competitive distance runners based on training age in a periodized macrocycle (4,10,36)**

|                                   | Day 1                                                                                                    |                                                                                                                  | Day 2                                                                                                         |                                                                                                                                | Day 3                                                                                                |                                                                                                         |
|-----------------------------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
|                                   | Exercise selection                                                                                       |                                                                                                                  | Exercise selection                                                                                            |                                                                                                                                | Exercise selection                                                                                   |                                                                                                         |
|                                   | Low training age                                                                                         | High training age                                                                                                | Low training age                                                                                              | High training age                                                                                                              | Low training age                                                                                     | High training age                                                                                       |
| GPP—<br>Mesocycle 1               | 1. Goblet squat<br>2. Split squat<br>3. Hamstring curl machine<br>4. KB swing<br>5. CMJ<br>6. Hurdle hop | 1. BB back squat<br>2. DB walking lunges<br>3. Nordic hamstring<br>4. Clean/rack pull<br>5. CMJ<br>6. Hurdle hop | 1. DL/RDL<br>2. SL squat<br>3. PF variation<br>4. High-effort box jump<br>5. Pogo jump<br>6. Drop/depth jumps | 1. DL/RDL<br>2. SL squat<br>3. PF variation<br>4. High-effort box jump<br>5. Pogo jump<br>6. Drop/depth jumps                  | 1. Leg press<br>2. SLDL<br>3. Machine knee extension<br>4. Jump squat<br>5. Bounding<br>6. Tuck jump | 1. BB back squat<br>2. SLDL<br>3. Bulgarian split squat<br>4. Jump squat<br>5. Bounding<br>6. Tuck jump |
| GPP—<br>Mesocycle 2               | 1. Goblet squat<br>2. Split squat<br>3. Hamstring curl machine<br>4. KB swing<br>5. CMJ<br>6. Hurdle hop | 1. BB back squat<br>2. DB walking lunges<br>3. Nordic hamstring<br>4. Clean/rack pull<br>5. CMJ<br>6. Hurdle hop | 1. DL/RDL<br>2. SL squat<br>3. PF variation<br>4. High-effort box jump<br>5. Pogo jump                        | 1. DL/RDL<br>2. SL squat<br>3. PF variation<br>4. High-effort box jump<br>5. Pogo jump                                         | 1. Leg press<br>2. SLDL<br>3. Machine knee extension<br>4. CMJ<br>5. Bounding                        | 1. BB back squat<br>2. SLDL<br>3. Bulgarian split squat<br>4. CMJ<br>5. Bounding                        |
| SPP—<br>Mesocycle 3               | 1. Squat variation<br>2. Hamstring curl<br>3. KB swing<br>4. CMJ<br>5. Hurdle hop                        | 1. BB back squat<br>2. Nordic hamstring<br>3. Clean/rack pull<br>4. CMJ<br>5. Hurdle hop                         | 1. DL/RDL<br>2. Split squat<br>3. Jump squats<br>4. High-effort box jump<br>5. Pogo jump                      | 1. DL/RDL<br>2. Bulgarian split squat or SL squat<br>3. Weighted BB 1/4 jump squats<br>4. High-effort box jump<br>5. Pogo jump | 1. Leg press<br>2. SLDL<br>3. KB swing<br>4. CMJ<br>5. Pogo jump                                     | 1. BB back squat<br>2. SLDL<br>3. Clean/rack pull<br>4. CMJ<br>5. Pogo jump                             |
| Maintenance<br>—<br>Mesocycle 4   | 1. Squat variation<br>2. Hamstring curl or SLDL<br>3. KB swing<br>4. CMJ<br>5. Hurdle hop                | 1. BB back squat<br>2. Nordic hamstring or SLDL<br>3. Clean/rack pull<br>4. CMJ<br>5. Hurdle hop                 | 1. DL/RDL<br>2. Split squat<br>3. Jump squats<br>4. High-effort box jump<br>5. Pogo jump                      | 1. DL/RDL<br>2. Bulgarian split squat or SL squat<br>3. Weighted BB 1/4 jump squats<br>4. High-effort box jump<br>5. Pogo jump | —                                                                                                    | —                                                                                                       |
| Taper/<br>peaking—<br>Mesocycle 5 | 1. Squat variation<br>2. Hamstring curl or SLDL<br>3. KB swing<br>4. CMJ<br>5. Hurdle hop                | 1. BB back squat<br>2. Nordic hamstring or SLDL<br>3. Clean/rack pull<br>4. CMJ<br>5. Hurdle hop                 | 1. DL/RDL<br>2. Split squat<br>3. Jump squats<br>4. High-effort box jump<br>5. Pogo jump                      | 1. DL/RDL<br>2. Bulgarian split squat or SL squat<br>3. Weighted BB 1/4 jump squats<br>4. High-effort box jump<br>5. Pogo jump | —                                                                                                    | —                                                                                                       |

BB = barbell; CMJ = countermovement jump; DB = dumbbell; DL = deadlift; GPP = general preparatory phase; PF = plantarflexion; RDL = Romanian deadlift; SL = single leg; SLDL = single-leg deadlift; SPP = specific preparatory phase.

**Table 4**  
Concurrent training prescription for resistance training sessions based on the periodized phase of training for competitive endurance runners

| Phase:                                                          | GPP                                                                  | GPP                                                                       | SPP                                                                                             | Maintenance                                                             | Taper/peak                                                          |
|-----------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------|
| Time frame:                                                     | Mesocycle 1 (4–6 wk)                                                 | Mesocycle 2 (4–6 wk)                                                      | Mesocycle 3 (4–6 wk)                                                                            | Mesocycle 4 (2–4 wk)                                                    | Mesocycle 5 (~2 wk)                                                 |
| Resistance training goal of the phase:                          | Induce neuromuscular adaptations and accumulate muscular strength    | Build maximal force development and accumulate muscular strength          | Improve rate of force development, maximal strength development, and increase peak power output | Preserve muscular strength and power output and avoid training loss     | Preserve muscular strength and power output and retain performance  |
| Maximal strength development exercise prescription:             | 3 exercises, 4–6 sets × 6–10 reps @ 70–80% 1RM, 2-min rest intervals | 3 exercises, 2–4 sets × 3–6 reps @ 75–85% 1RM, 2- to 3-min rest intervals | 3 exercises, 2–4 sets × < 5 reps @ >80% 1RM, 3 + min rest intervals                             | 3 exercises, 1–2 sets × < 5 reps @ >80% 1RM, 2- to 3-min rest intervals | 3 exercises, 1–2 sets × < 5 reps @ 50–60% 1RM, 2-min rest intervals |
| Explosive strength and reactive strength exercise prescription: | Up to 6 exercises, total of 60–100 contacts                          | 3–4 exercises, 100–200 contacts                                           | 3–4 exercises, 100–200 contacts                                                                 | 3–4 exercises, 50–100 contacts                                          | 3–4 exercises, 50–100 contacts                                      |

GPP = general preparatory phase; reps = repetitions; RM = repetition maximum; SPP = specific preparatory phase.

gradually increase in volume and intensity. Moderate-distance and moderate-effort runs will build exercise capacity and conditioning while allowing the athlete to recover from the last peaking phase. Any gait alterations, drills, or changes in gait pattern should be introduced in this phase. Variability in running surfaces, inclines, percentage of race pace, and distances will be high in this phase. When runners enter the SPP, they should have a strong foundation of conditioning and will begin to transition to greater sessions that are specific to their individual race distance, pace, and surface. The hallmarks of transition from GPP to SPP are a decrease in variability, increase in specificity, decrease in volume, and increase in intensity of runs that will continue until the peaking phase. In the taper/peaking phase, individual differences between runners including rates of recovery will dictate the magnitude of deload that will occur.

The prescription of endurance running training intensity distribution is debated

in the literature, and variables for determining training zones are highly diverse in research and in practice. Physiological factors such as LT, ventilatory threshold, percentage of  $\dot{V}O_{2\max}$ , and percentage of maximum heart rate, and subjective factors such as session rating of perceived exertion are commonly used. Recommendations for the intensity distribution of endurance training sessions in this review will be based on the race-paced zones 1 through 3; however, session rating of perceived exertion measures may be an additional valuable method to track subjective fatigue measures (20).

Race-paced zones and session rating of perceived exertion, a method based on the individual perception of effort, are easy to use in practice with a wide range of runners and can be tracked easily. A study by Manzi et al. (27) supports the use of session rating of perceived exertion due to its ease of use in practice and large correlations with 2 standardized, validated objective measures based on heart rate. A review examining training

intensity distribution by Kenneally et al. in 2018, recommends that zones based on race pace offer an easily understandable method of programming intensity and a compatible way to compare endurance training distributions. This review provides evidence supporting polarized and pyramidal training intensity distributions with a high volume of low-intensity training and a low volume of high-intensity training based around race-paced zones for long-distance runners. The pyramidal approach, in which 80% of training occurs in a low-intensity zone and the remaining 20% is split in a decreasing manner between a moderate-intensity zone and a high-intensity zone, and a polarized approach, in which approximately 80% of training occurs in a low-intensity zone and approximately 20% of training occurs in the high-intensity zone with little or none occurring in the moderate-intensity zone, are the 2 most popular training zone distributions for endurance runners and the 2 most studied distributions (23). Notably, threshold training



consistently proves to be less effective in peer-reviewed evidence; however, there is anecdotal evidence promoting training in higher volumes (>20%) in the moderate-intensity zone (23).

The zones for the race-paced approach were determined in the review as follows: zone 1, volume performed <95% of goal race pace; zone 2, volume performed between 95 and 105% of goal race pace; and zone 3, volume performed at >105% of goal race pace (23). Table 2 offers an example of how to implement these race-paced zones of training with resistance training sessions in a periodized macrocycle for endurance runners.

In an observational study of marathoners and ultramarathoners, Tanda states that training speed (pace) was the main variable predicting 42-km race performance. Training volume was found to be the main factor predicting 100-km race times (42). Kenneally's review reflects a similar message in which shorter races require greater volume of training to be performed at faster paces than race pace, and longer races require increasing volumes in and around race pace (23). These conclusions may suggest that strength and conditioning practitioners program a greater number of sessions in zones 2 and 3 for 10,000 m to half-marathon distance runners and a greater number of sessions in zone 2 for marathoners, and focus on greater overall volume of running training in zone 1 for ultramarathoners. Mindfully, coaches and athletes must take excessive running volume into consideration with the addition of concurrent resistance training due to the acute interference effects of residual muscular fatigue and the potential for overtraining and injury (36).

## CONCLUSION

Concurrent resistance training in competitive distance runners is recommended due to the positive performance effects in RunE. Negligible changes in body composition will be observed with the addition of resistance training sessions. High-load

resistance training, explosive strength, and reactive strength resistance training will result in the greatest changes in RunE. Implementation of concurrent training in competitive distance runners should be programmed intelligently and with considerations for periodization, exercise selection, fatigue, current training status, and other training variables documented in this review.

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